Etienne Lee

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EDUCATION

University of Pennsylvania | GPA: 4.0/4.0 Expected May 2027

- Bachelor of Science in Engineering in Computer Science
- Master of Science in Engineering in Data Science

WORK EXPERIENCE

IceKredit | Data Science Intern June 2026 – August 2026

- Led a cross-domain, cross-lingual Alzheimer's detection project, first establishing an in-domain baseline (0.82 BA) with a fine-tuned HuBERT + RoBERTa pipeline before tackling generalization to a sparse, code-mixed target dataset
- Justified 0.70 balanced accuracy as a competitive, publishable target for this compounded setting via literature review, then achieved it through targeted domain-adaptation strategies and data augmentation

Conversion Capital | Data Science Intern June 2025 – August 2025

- Designed a data enrichment pipeline connecting Notion, Affinity, and Specter to automatically update investment signals across 3,000+ sourced companies, surfacing weekly top prospects for the investment team
- Built a bidirectional sync system between Excel and Notion using Python, eliminating manual data entry

Neopets | Software Engineering Intern June 2024 – August 2024

- Spearheaded development of a full-stack browser extension and a multiplayer game demo using React, TypeScript, Node.js, and Socket.IO
- Developed interactive frontend animations and dynamic UI components in React, alongside backend systems handling user authentication, real-time game state, and achievement tracking via Node.js and MongoDB

University of Pennsylvania | Teaching Assistant, CIS 5500: Databases August 2025 – Present

- Created homework and exam questions, mentored project groups, and held office hours for 200+ students covering SQL, database design, query optimization, and full-stack development

PROJECTS

PennOS (C, Bash)

- Co-developed a UNIX-like guest OS featuring a priority scheduler, FAT filesystem, and interactive shell
- Built and integrated filesystem with kernel from scratch, enabling file operations and UNIX commands (cat, mv, ls, etc.)

Ocean Biodiversity & Climate Change Explorer (React, Node.js, PostgreSQL, AWS)

- Ingested and filtered a 136M-row dataset down to a normalized 2M-row PostgreSQL schema using chunked processing
- Cut key query latency up to 4x via materialized views, precomputed spatial/depth indices, and targeted indexing

New Tab Photo Extension (TypeScript, React)

- Published and deployed a Chrome extension to the Web Store, built end-to-end using Chrome Storage API across photo upload, deletion, and randomized selection

News Source Classification (Python, PyTorch, HuggingFace Transformers)

- Scraped and expanded a Fox News vs. NBC News headline dataset to 17.9K samples for binary classification
- Fine-tuned RoBERTa/DeBERTa, achieving 91.75% leaderboard accuracy with a LinearSVC head

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript/TypeScript, HTML/CSS, SQL, Bash, RISC-V Assembly, OCaml
Frameworks: React, Node.js, Next.js, Scikit-learn, PyTorch
Tools: Git, MongoDB, PostgreSQL, Neo4j, AWS, Postman, Google Colab, LaTeX, Docker

LEADERSHIP AND HONORS

- **Hong Kong Junior National Tennis Champion** (2022); Member, Hong Kong National Tennis Squad (2016 – 2023)
- **HKSES (Hong Kong Scholarship for Excellence Scheme) Recipient** (2023)
- **First Place (Best Overall),** Penn NSF HDR ML Hackathon (2024)
- **Founder,** Tennis SENsation (2022 - Present): Led weekly tennis program for children with special needs